39 The table gives information about some of the planets in our Solar System.

planet	mass compared with the Earth	average density / g/cm^3	gravitational field strength at surface / N / kg
Earth	1.0	5.5	9.8
Jupiter	320	1.3	25
Mercury	0.060	5.4	3.8
Uranus	15	1.3	10

Which statement about these planets is correct?

A As the mass compared with the Earth increases, the average density increases.

B As the mass compared with the Earth increases, the gravitational field strength increases.

C As the average density increases, the gravitational field strength decreases.

D The average density is directly proportional to the gravitational field strength.